

Problem 2.20

You are given that $A(t) = at^2 + bt + c$, for $0 \leq t \leq 2$, and that $A(0) = 100$, $A(1) = 110$, and $A(2) = 136$. Determine $\delta_{\frac{1}{2}}$.

Solution

Since

$$A(t) = at^2 + bt + c,$$

we use the given values to determine $A(t)$.

From $A(0) = 100$, we get

$$c = 100.$$

From $A(1) = 110$, we get

$$a + b + 100 = 110,$$

so

$$a + b = 10.$$

From $A(2) = 136$, we get

$$4a + 2b + 100 = 136,$$

so

$$2a + b = 18.$$

Subtracting,

$$(2a + b) - (a + b) = 18 - 10,$$

which gives

$$a = 8.$$

Then

$$b = 2.$$

Therefore,

$$A(t) = 8t^2 + 2t + 100.$$

Now compute the accumulation function:

$$a(t) = \frac{A(t)}{A(0)} = \frac{A(t)}{100}.$$

Hence,

$$a(t) = \frac{8t^2 + 2t + 100}{100} = 0.08t^2 + 0.02t + 1.$$

Differentiate:

$$a'(t) = 0.16t + 0.02.$$

Recall that

$$\delta_t = \frac{a'(t)}{a(t)}.$$

Thus,

$$\delta_t = \frac{0.16t + 0.02}{0.08t^2 + 0.02t + 1}.$$

Evaluating at $t = \frac{1}{2}$,

$$\delta_{1/2} = \frac{0.16(0.5) + 0.02}{0.08(0.5)^2 + 0.02(0.5) + 1}.$$

Therefore,

$$\delta_{1/2} = \frac{0.10}{1.03} \approx 0.097087.$$

$$\boxed{\delta_{1/2} \approx 0.097087.}$$